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Chromosomal Aberrations

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Cytogenetics

- Cytogenesis is the study of chromosomes structure and number and their related disease states
- Chromosomes are complex structures located in the cell nucleus, they are composed of DNA, histone and non-histone proteins, RNA, and polysaccharides.
- They are basically the "packages" that contain the DNA.

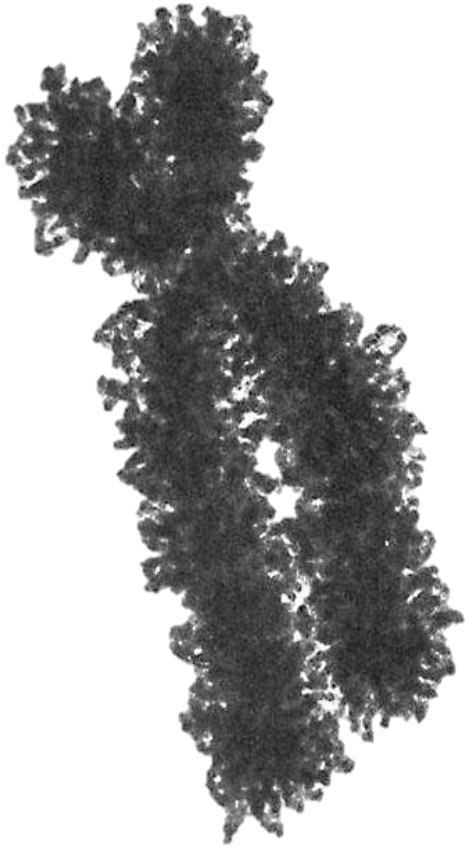


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Cytogenetics

- Normally chromosomes can't be seen with a light microscope but during cell division they become condensed enough to be easily analyzed at 1000X.
- To collect cells with their chromosomes in this condensed state, they are exposed to a **mitotic inhibitors** which blocks formation of the spindle and arrests cell division at the metaphase stage.



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Cytogenetics

- A variety of tissue types can be used to obtain chromosome preparations - peripheral blood, bone marrow, amniotic fluid, and products of conception.
- Although specific techniques differ according to the type of tissue used, the basic method for obtaining chromosome preparations is as follows:
  1. Sample log-in and initial setup.
  2. Tissue culture (feeding and maintaining cell cultures).
  3. Addition of a mitotic inhibitor to arrest cells at metaphase.
  4. Stain chromosome preparations to detect possible numerical and structural changes.

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Staining

- To identify different regions of the chromosomes
- Gives different types of bands and inter-bands

Types of Staining

Q- Banding

- Dye – Quinacrine / UV light
- Chromatids are stained, except the distal ends

G-Banding

- Geimsa stain
- Same as above – no florescent microscopy required

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Staining

C-Banding

- Constitutive heterochromatin
- Centromere region

R-Banding

- Acridine orange
- Reverse banding / all areas which lie between Q-bands

Significance

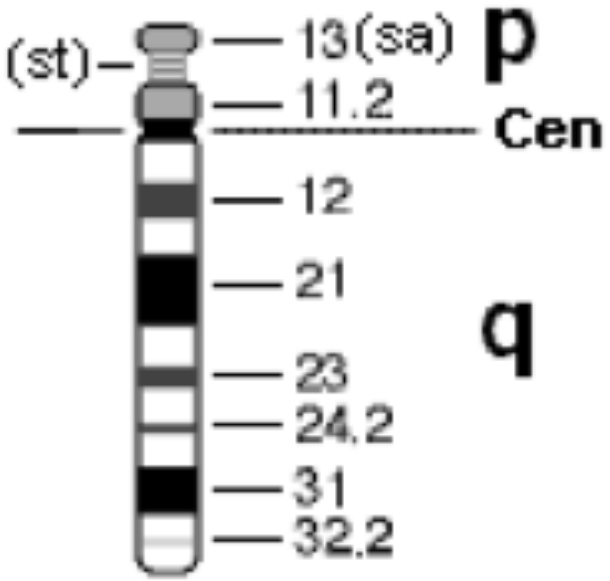
- Chromosomal analysis helped correlating a number of diseases to structural deformities and numerical abnormalities

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Ideograms

- The ideogram is basically a "chromosome map" showing the relationship between the short and long arms, centromere (*cen*), and in the case of acrocentric chromosomes the stalks (*st*) and satellites (*sa*).
- The specific banding patterns are also illustrated. Each band is numbered to aid in describing rearrangements.



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Chromosomal Abnormalities

- Although chromosome abnormalities can be very complex there are two basic types: ***numerical*** and ***structural***.
- Both types can occur simultaneously.

Numerical Abnormalities

- Numerical abnormalities involve the loss and/or gain of a whole chromosome/s and can include both autosomes and sex chromosomes.
- Generally chromosome loss has a greater effect on an individual than does chromosome gain, although these can also have severe consequences.

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Numerical Abnormalities

- Cells which have lost a chromosome show ***monosomy*** for that chromosome while those with an extra chromosome show ***trisomy*** for the chromosome involved.
- Nearly all autosomal monosomies die shortly after conception and only a few trisomy conditions survive to full term.
- Curiously, a condition called ***triploidy*** in which there is an extra copy of every chromosome (69 total), can occasionally survive to birth, but usually die in the newborn period.
- Another general rule is that loss or gain of an autosome has more severe consequences than loss or gain of a sex chromosome.

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Structural Abnormalities

- **Structural abnormalities** involve changes in the structure of one or more chromosomes.
- They can be incredibly complex but our focus will be on the three of the more common types:
  - Deletions
  - Duplication
  - Inversions
  - Translocations

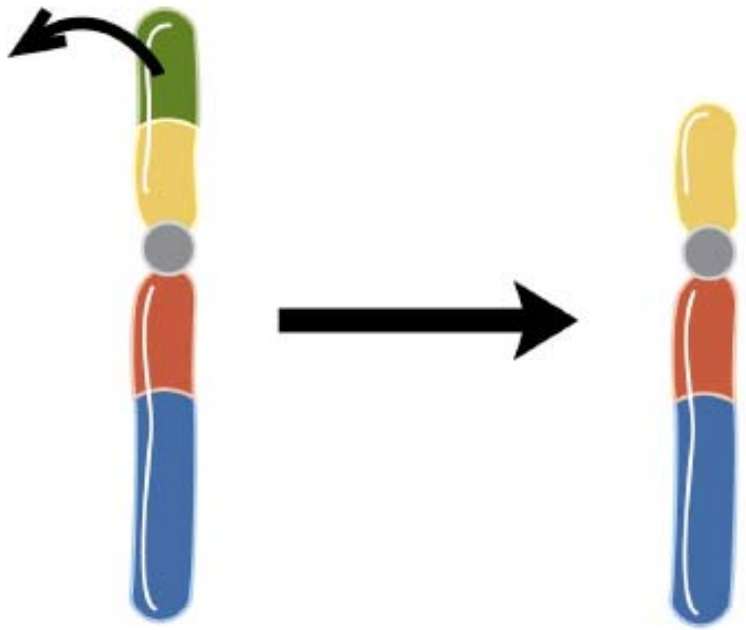
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Structural Abnormalities

Deletions

- Involve loss of material from a single chromosome, a fragment without centromere
- The effects are typically severe and lethal since critical genes will be missing



Deletion

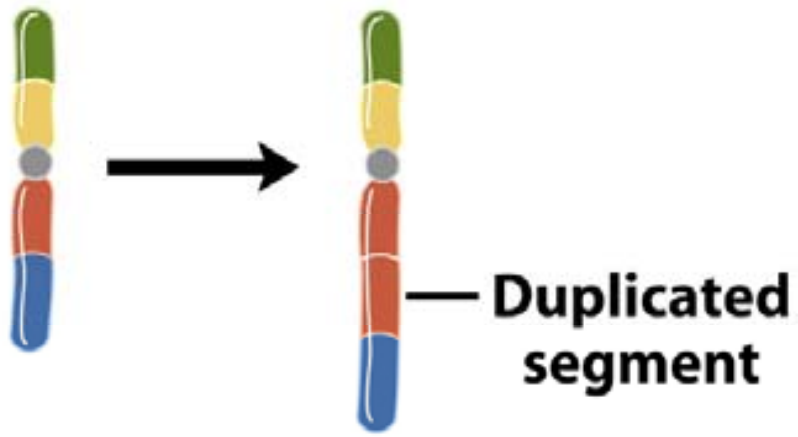
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Structural Abnormalities

Duplication

- A deleted chromosome fragment can attach to its homologue, thereby duplicating a region of genes
- In some cases, a segment of a gene or chromosome undergoes multiple repetitions.



Duplication

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Structural Abnormalities

Gene Transfers

- Normally recombination between homologous chromosomes during meiosis, changes specific alleles on a given chromosome and adds to variation in a population
- But in gene transfer, two things can happen
  - Genes from one chromosome get transferred to another chromosome
  - Genes can get rearranged on the same chromosome
- There are two types of gene transfers – Inversions & Translocations (or transpositions)

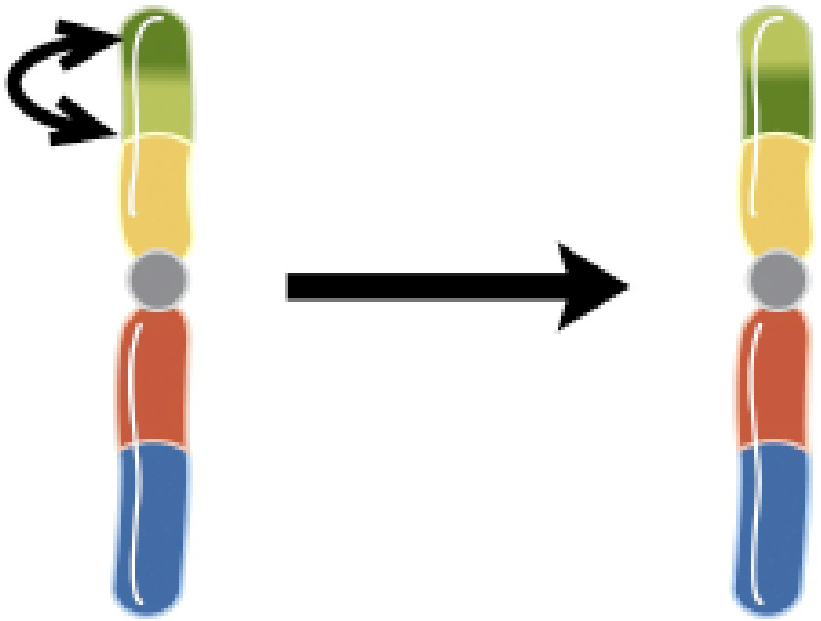
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Structural Abnormalities

Gene Inversions

- For an inversion, a group of genes can have their order reversed in the chromosome so that a gene sequence that should be **A-B-C-D-E-F-G-H** is changed to **A-B-F-E-D-C-G-H** instead.
- This results in part of the DNA being "backwards" and unreadable.
- Many gametes are not viable after inversions.



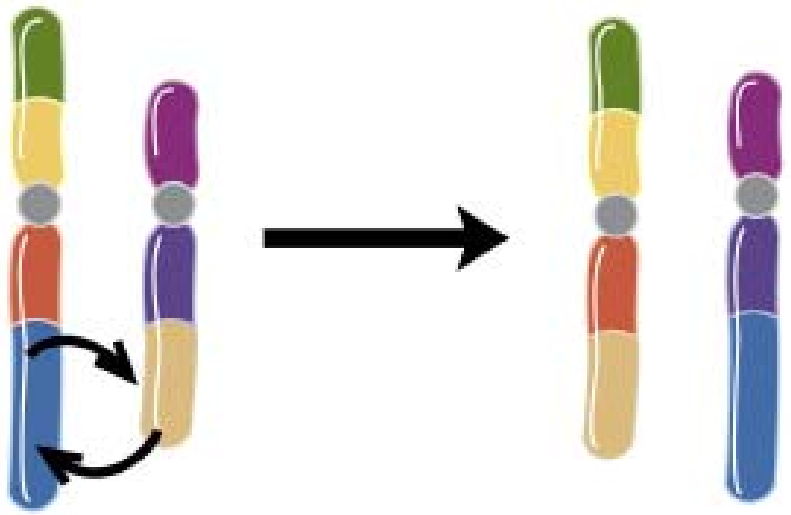
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Structural Abnormalities

Gene Translocations

- A gene or part of chromosome can be transposed or transferred from one part of the chromosome to another or from one chromosome to another non-homologous chromosome.
- **Reciprocal or balanced** translocations involve exchange of genes between non-homologous chromosomes.
- **Transposons** – gene elements that move / also **called jumping genes - Barbara McClintock.**

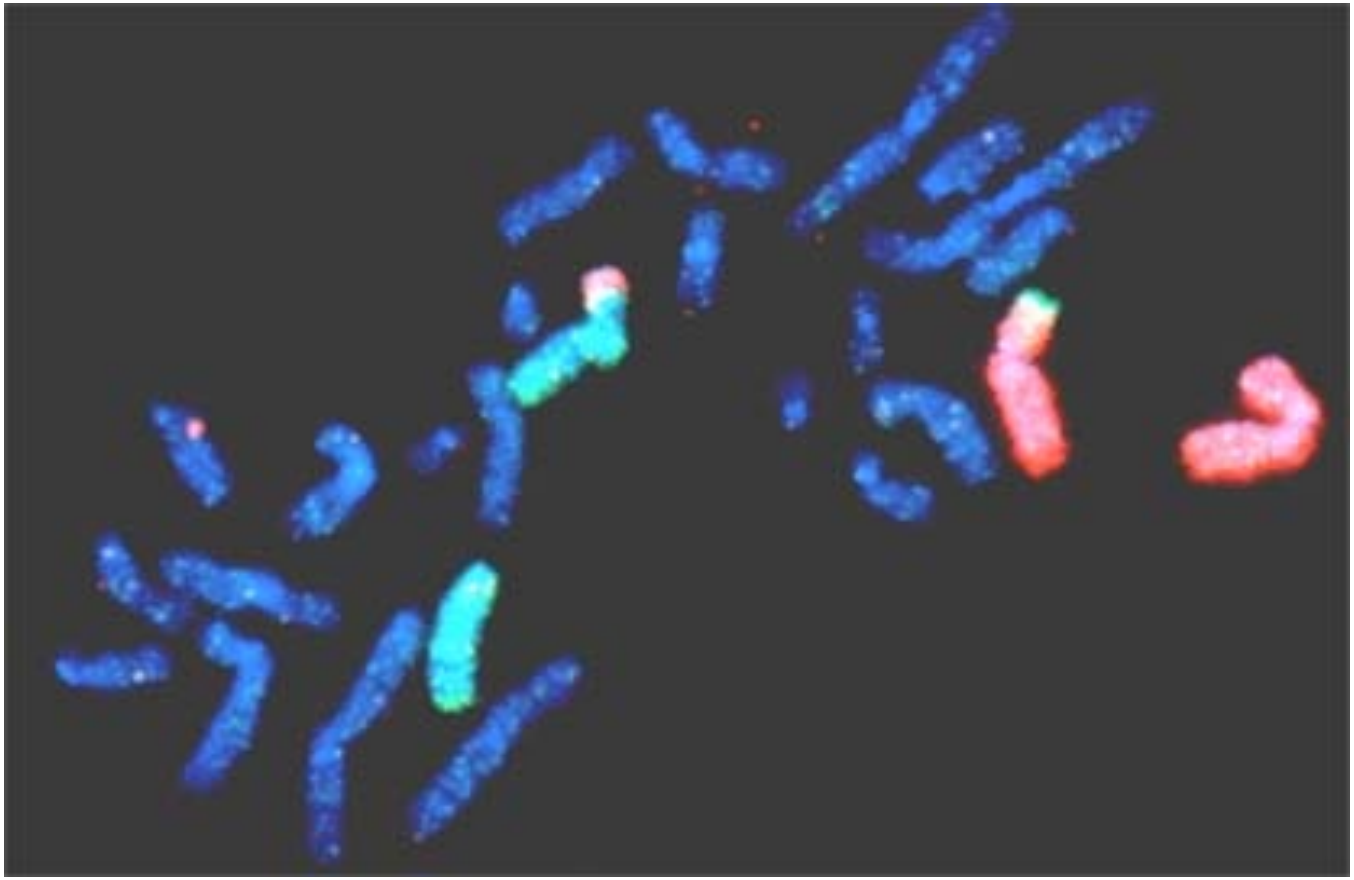


Translocation

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Translocations



Red and Green illuminated chromosomes illustrate translocation.

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Chromosomal Aberrations

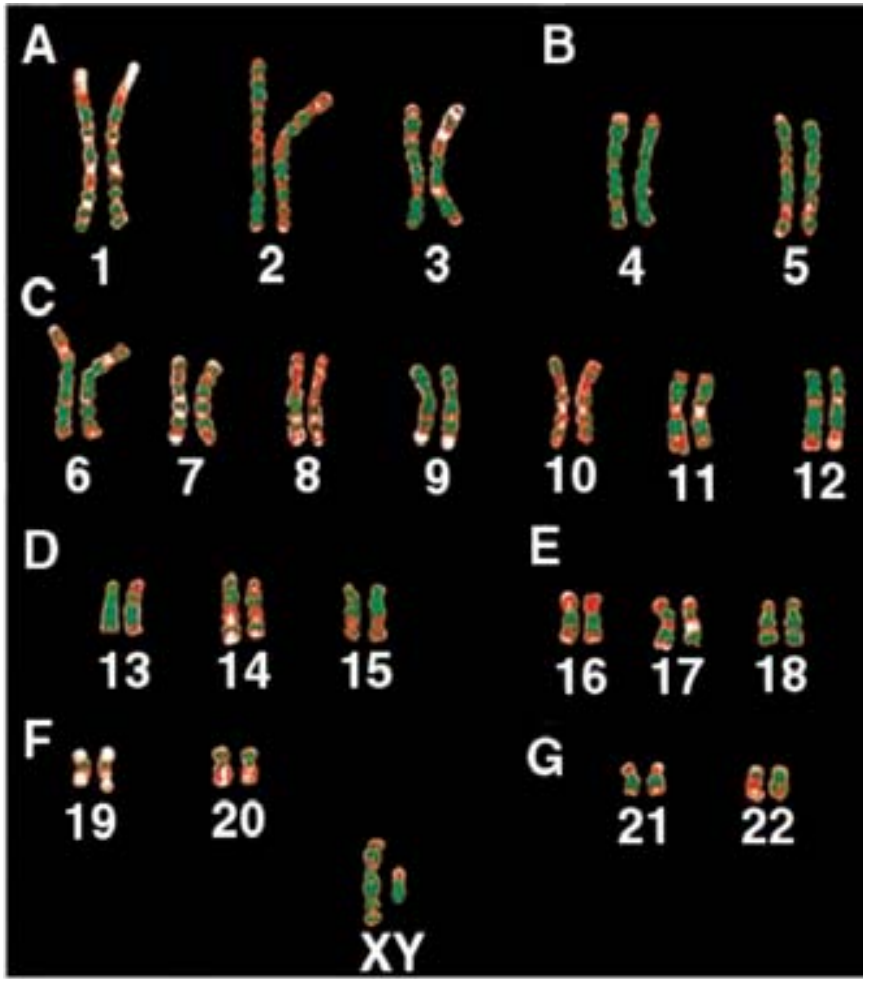
- Both numerical and structural aberrations can also be classified into
  - **Constitutional –** Those are born with - Familial
  - **Acquired –** Those that arise as secondary changes due to other diseases like cancer – De Novo
- **Mosaicism**
  - Sometimes individuals are found with both normal and abnormal cell lines
  - Mosaics are usually found in abnormal numbers and structural mosaics are very rare
  - Hence, any routine cytogenetic analysis should involve 15-20 cells to rule out clinically significant mosaicism

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Numerical Abnormalities

- Easy to observe 46 chromosomes and their shapes by obtaining a karyotype or chromosome display through various staining techniques
- Any abnormalities in chromosomal number can be easily detected



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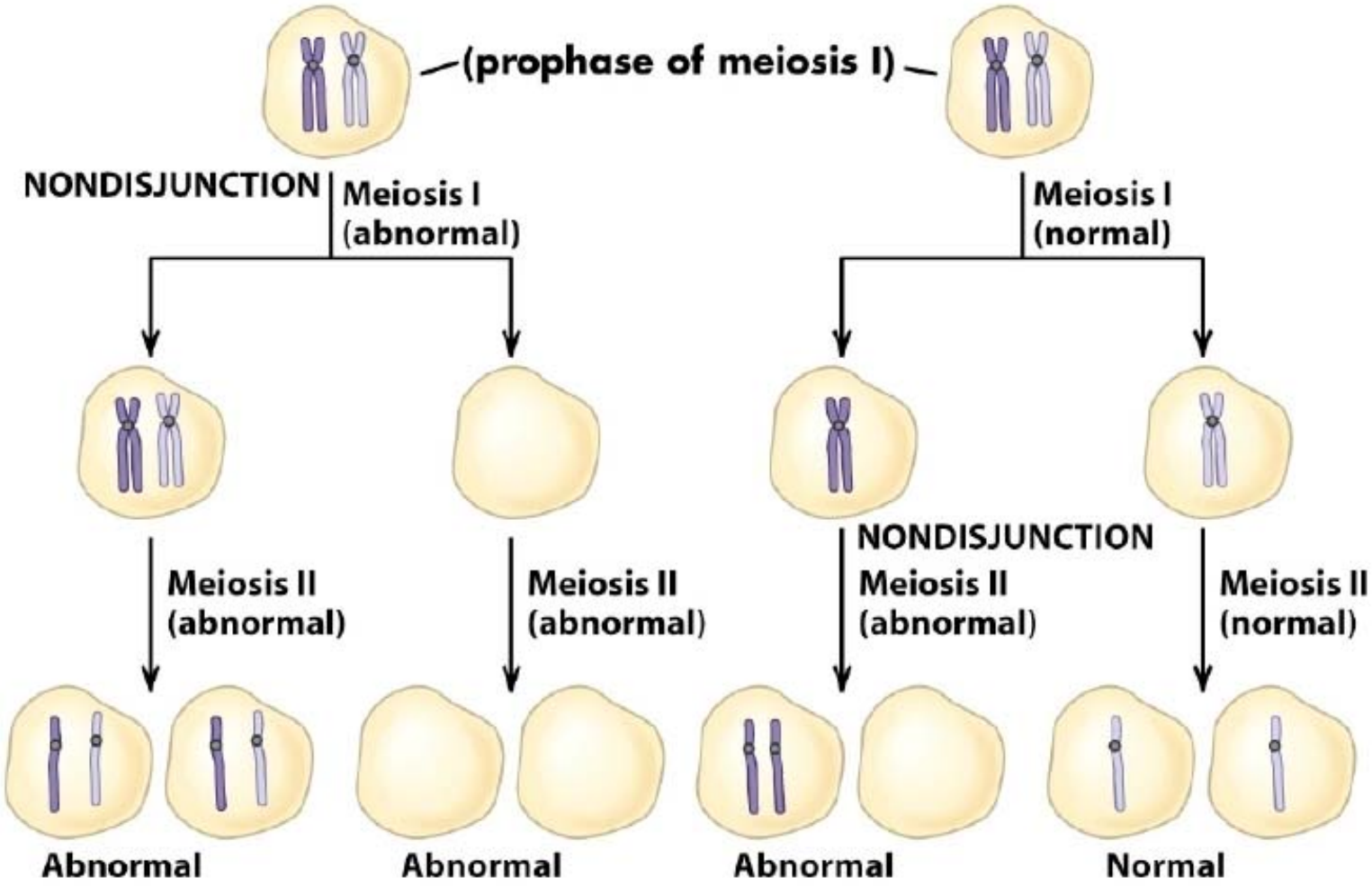
Nondisjunction & Numerical Aberrations

- Normal chromosomal numbers are haploid in gametes and diploid in zygotes – Euploidy (true number)
- Any deviation from this is called **Aneuploidy** when one chromosome is involved and **Polyploidy** when the entire set of chromosomes are involved like triploidy.
- For reasons not fully understood, occasionally, a homologous chromosome pair will fail to separate during meiosis
- This results in an egg or sperm with one more or one less than the normal complement of chromosomes (trisomy or monosomy).
- This is referred to as **non-disjunction,** which can occur either in the first or second meiotic divisions

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Nondisjunction



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Nondisjunction

- Most often, a non-disjunction results in a gamete that does not survive
- In some cases, however, some gametes do survive, producing individuals with abnormal chromosome numbers.
- Most of these non-disjunctions have serious genetic consequences.
- A non-disjunction can affect either the sex chromosomes or autosomes.

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Down's Syndrome

- Named after Langdon Down (1886)
- Trisomy 21 or extra copy of chromosome 21
- Hence the person has 47 instead of normal 46 chromosomes
- Karyotype represented as **(47, +21)**

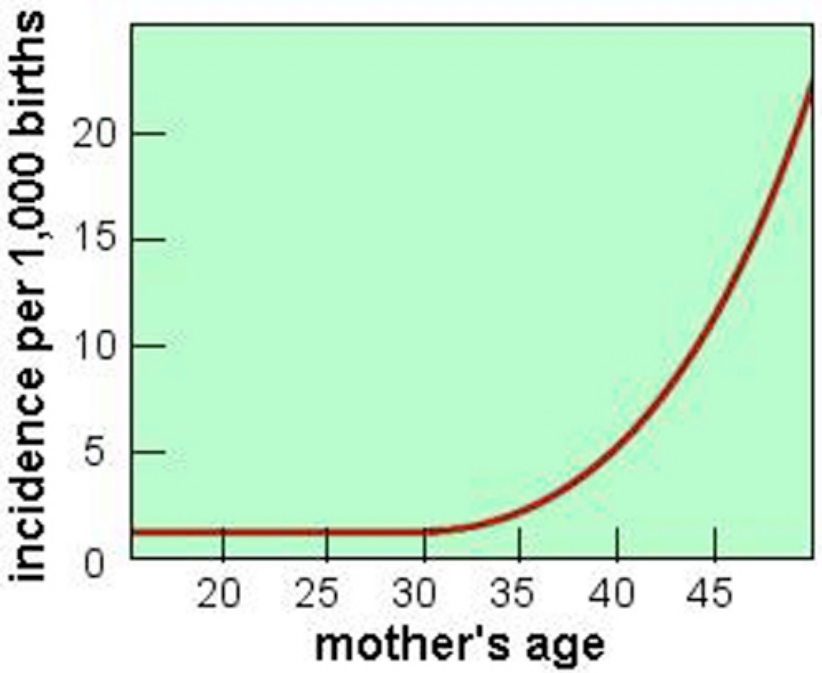


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Down's Syndrome

- As many as 1 in 20 eggs produced after the age of 40 may carry this chromosome abnormality.
- Because nondisjunction seems to be common in women who are aged
- The incidence of Down's Syndrome drastically increases with age of mothers



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Down's Syndrome

- Presence of Simian Crease in the palm
- Tendency for every finger to have a loop in the print region
- Short stature (4 ft)
- Large tongues with distinctive furrowing
- Loose jointed appendages
- Mental retardation
- Defects in cardio-vascular system - short life expectancy
- Epicanthic fold

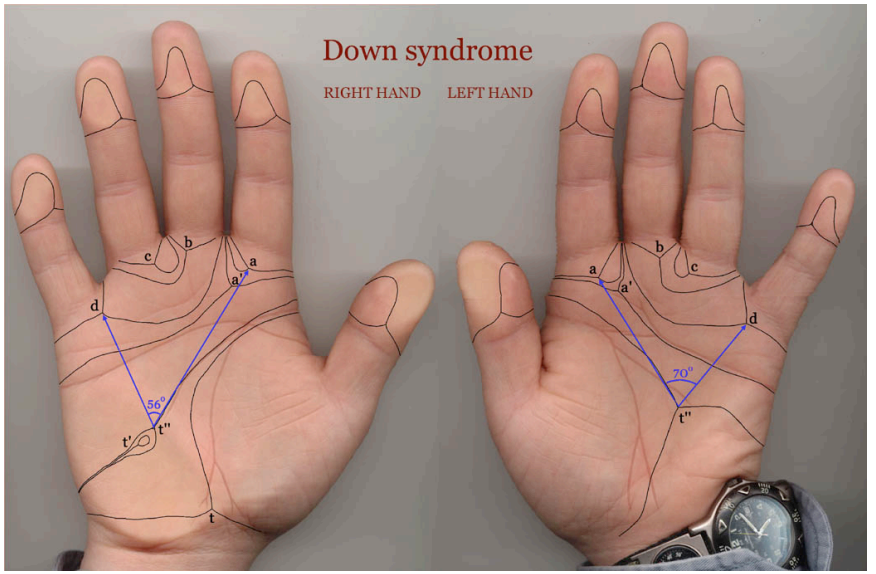


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Down's Syndrome







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Patau Syndrome

- Trisomy 13 (47, +13)
- Can also be mosaicism
- 1 out of every 10,000 newborns

Symptoms

- Cleft lip or palate
- Clenched hands (with outer fingers on top of the inner fingers)
- Close-set eyes -- eyes may actually fuse together into one
- Decreased muscle tone



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Patau Syndrome

- Extra fingers or toes (polydactyly)
- Hole, split, or cleft in the iris
- Low-set ears
- Mental retardation, severe
- Seizures
- Single palmar crease
- Skeletal (limb) abnormalities
- Small head (microcephaly)
- Small lower jaw (micrognathia)



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Edward Syndrome

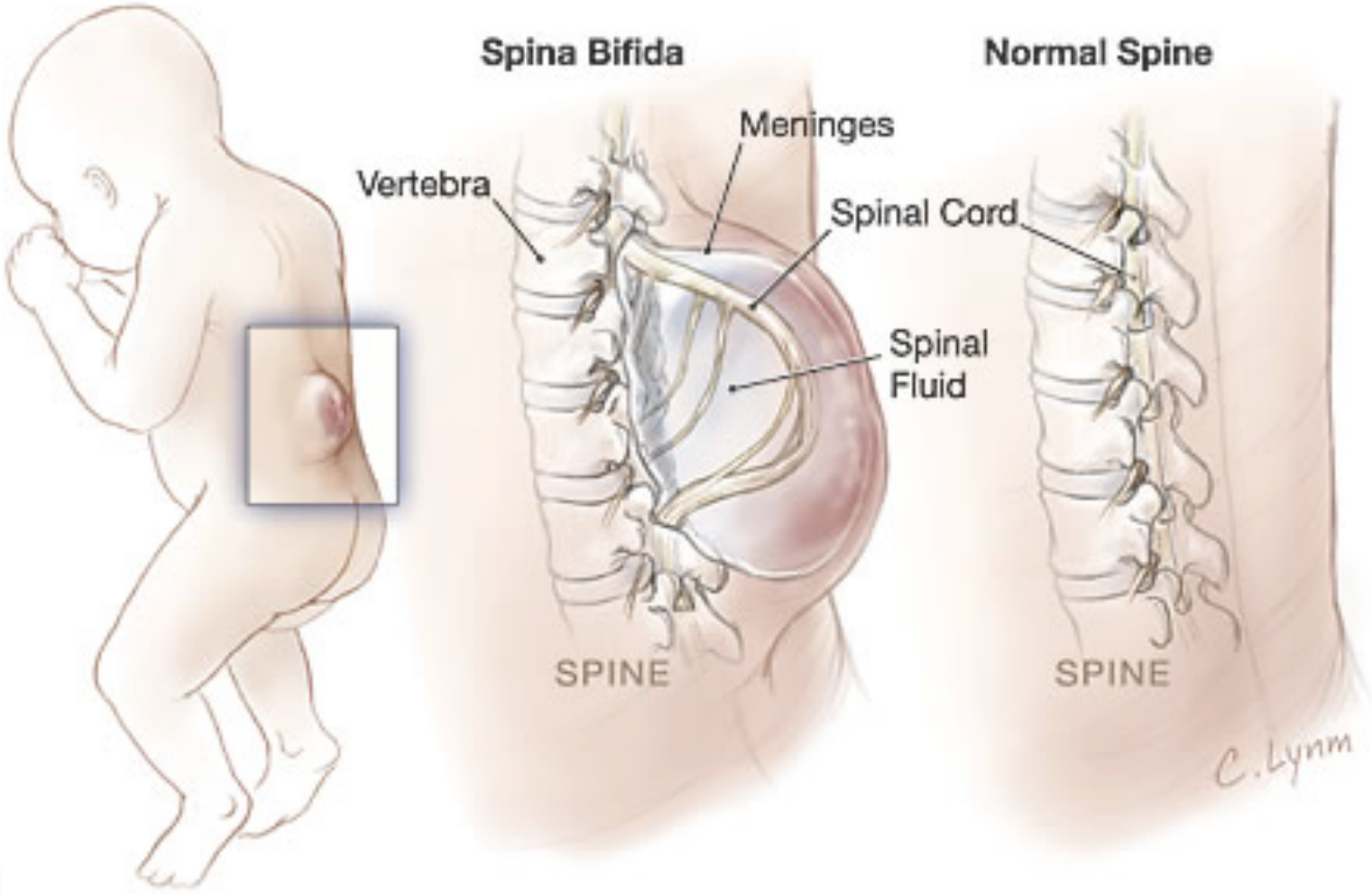
- Trisomy 18 (47, +18)
- Facial deformities
- Fingers curled over one another in clenched fists
- Heart defects kidney abnormalities
- Low, small ears
- Microcephaly (small head and BRAIN)
- Small mouth and cleft deformities
- Spina bifida
- Syndactyly



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Spina Bifida



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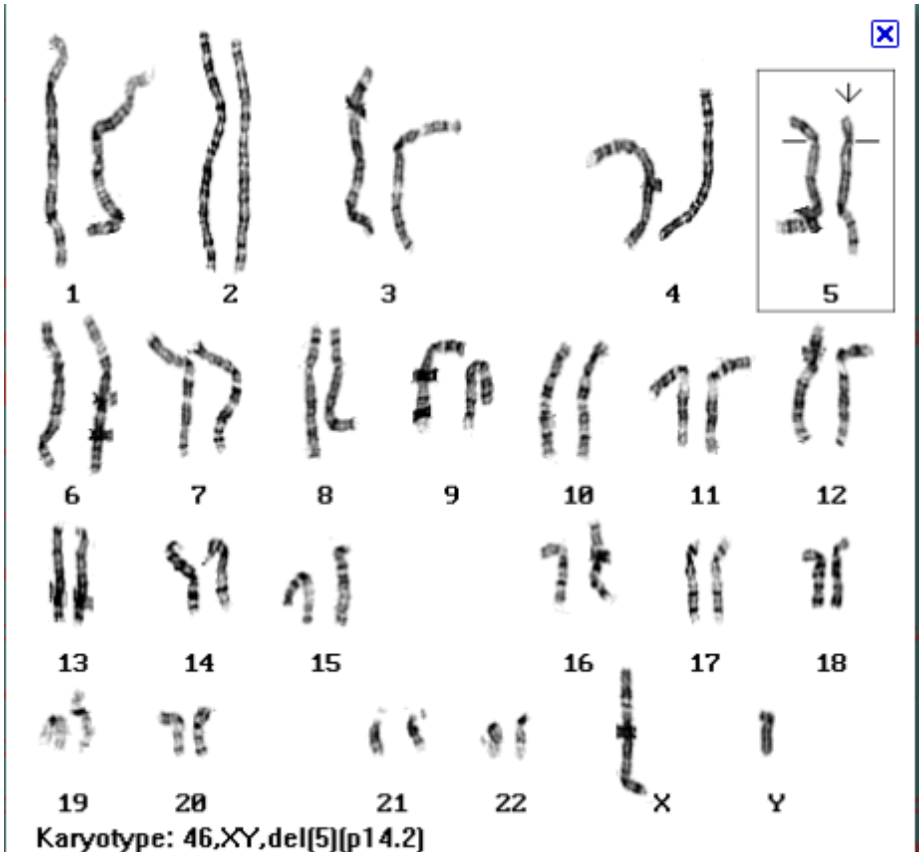
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Cri Du Chat Syndrome

- Deletion aberration on Chromosome on p-arm
- Karyotype (46, XX) 5p –

Symptoms

- Cry that is high-pitched and sounds like a cat
- Downward slant to the eyes
- Low birth weight and slow growth
- Low-set or abnormally shaped ears



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Cri Du Chat Syndrome

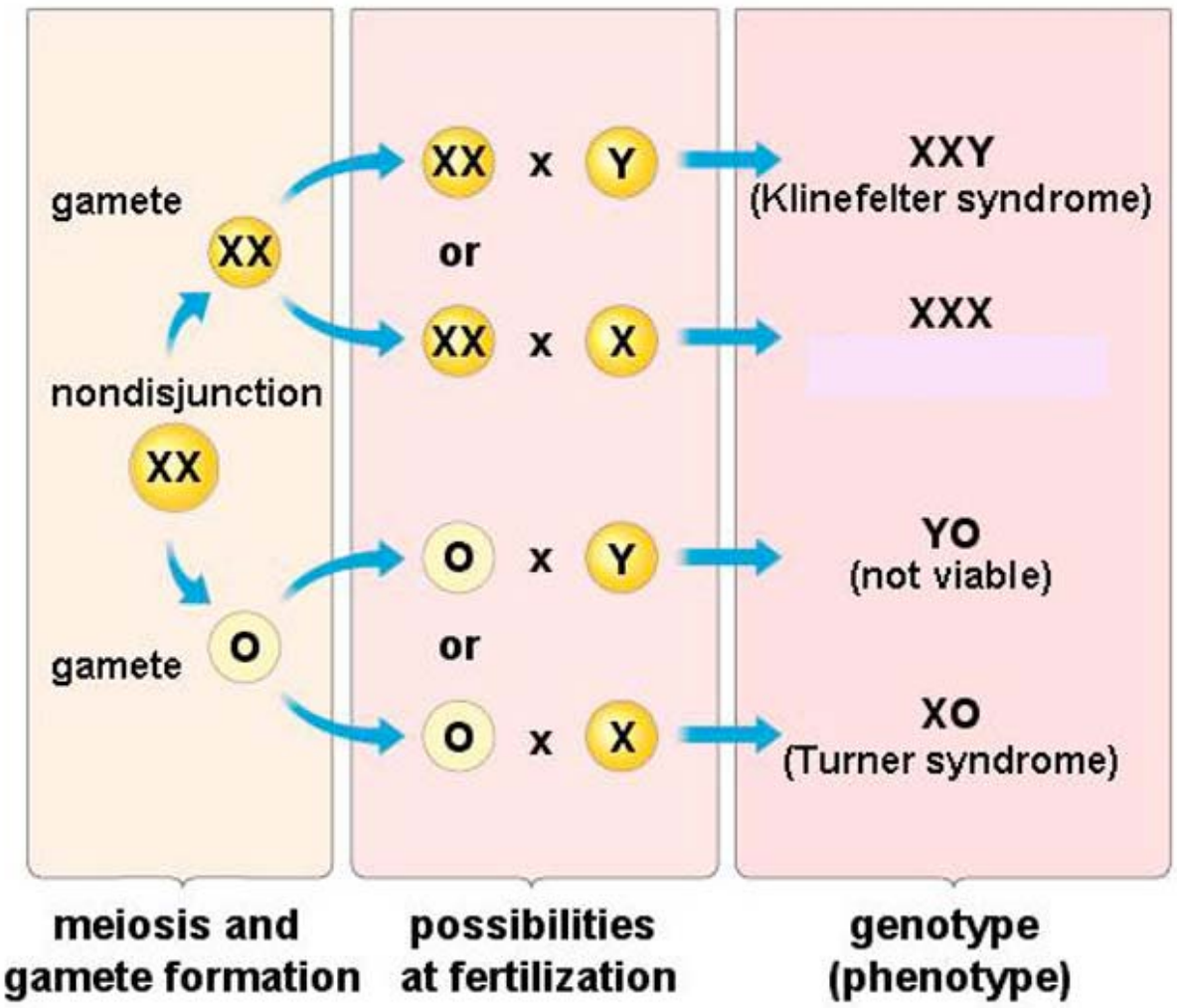
- Mental retardation
- Partial webbing or fusing of fingers or toes
- Single line in the palm of the hand (simian crease)
- Slow or incomplete development of motor skills
- Small head (microcephaly)
- Small jaw (micrognathia)
- Wide-set eyes



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Sex Chromosome Aberrations



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Turner's Syndrome

- Monosomy XO (45, X)
- One out of 2500 female births
- Ullrich-Turner syndrome / Gonadal Dysgenesis
- Ovaries are absent
- Limited secondary sexual characters
- Osteoporosis
- Sterility
- Short stature and fingers
- Webbed neck
- Many internal abnormalities in cardiovascular and renal systems

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Klienfelter Syndrome

- 47, XXY
- Abnormal body proportions (long legs, short trunk, shoulder equal to hip size)
- Abnormally large breasts (gynecomastia)
- Infertility
- Sexual problems
- Less than normal amount of body hair
- Tall height

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Triple X Syndrome

- 47, XXX / Super Females
- Increased risk of learning disabilities and delayed development of speech and language skills.
- Delayed development of motor skills (such as sitting and walking),
- Weak muscle tone (hypotonia), and behavioral and emotional difficulties
- Seizures or kidney abnormalities occur in about 10 percent of affected females.

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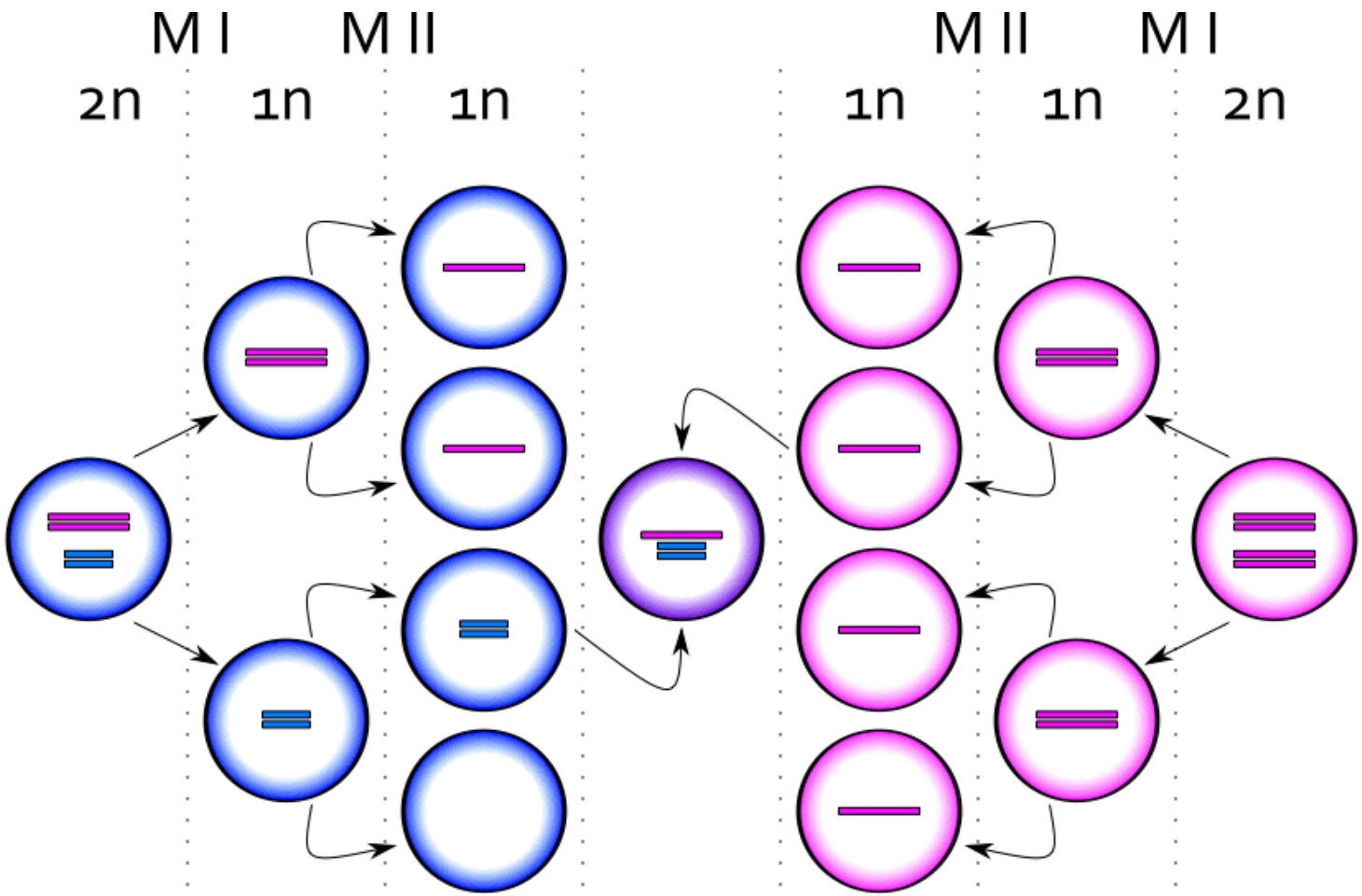
XYY / Criminal Syndrome

- 47, XYY
- Originally called criminal syndrome – now known more as Jacob Syndrome
- 97% of men do not know their karyotype as the phenotype is “normal”
- Usually detected only during a genetic analysis
- 50% of boys show risks of learning disabilities, language acquisition and delayed speech
- Tall statures

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XYY Nondisjunction



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Extra Nuclear Inheritance

- Mendelian inheritance addresses the behavior of genes on chromosomes.
- Mitochondria have small circular pieces of DNA and that DNA is transcribed and translated within the organelle.
- Mitochondria are self-replicating.
- In sexual reproduction, only the egg cell's cytoplasm is passed to the zygote, so only maternal mitochondrial DNA will be transmitted from generation to generation.
- Some genetic disorders are traced to mutations in mitochondrial DNA that codes for proteins
- Mutations in mitochondrial DNA may be one reason cells age.

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Intersex States

- Conditions in which the appearance of the external genitalia (genital /sex organs) is either ambiguous or at variance with the person's chromosomal or gonadal sex (testes-male, ovaries-female)

Etiology and Classification

- Genitalia form during the first 3 months of gestation via a cascade of events initiated by the fetal karyotype and mediated largely by sex steroids
- Aberrations in this cascade produce genital ambiguities or inconsistencies resulting in intersex states
- Classification is based on gonadal histology

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Intersex States

Female pseudohermaphrodites

- Have ovaries and normal female internal genitalia but ambiguous external genitalia
- They are genetically normal females with karyotype 46,XX
- The ambiguous external genitalia result from exposure to excessive amounts of androgens in utero (inside uterus)
- The offending androgen may be
  - Exogenous – progesterone given to mother prevent miscarriage)
  - Endogenous – enzymatic block due to aberrations in chromosome 6 (adrenal virilism)

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Intersex States

Male pseudohermaphrodites

- Have gonadal tissue that is only testicular and usually have 46, XY karyotype
- External and internal genitalia are ambiguous – a lot of variability in phenotypes
- In some cases a female phenotype is seen in complete form – testicular feminization syndrome
- In some other cases testes are seen with presence of uterus
- Etiology – can happen because of inadequate production of androgens or inadequate response to androgens during critical growth phase

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Intersex States

True Hermaphrodites

- Have both ovarian and testicular tissue and mixed masculine and feminine genital structures
- Either the ovarian or testicular tissue predominates
- Most of true hermaphrodites have karyotype 46,XX
- Rarely, in them, external genitalia is fully masculinized

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Genetic Imprinting

Also called Genome Imprinting

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Theory of Equivalence

- ***There is no difference in the phenotype of the trait even in case of reciprocal cross***
- Reciprocal cross - an exchange of sex of parents carrying different genotypes
- Theory of equivalence is NOT UNIVERSAL
  - Differs from gene to gene in same animal
  - Differs from animal to animal
- An exception to this theory of equivalence is GENETIC IMPRINTING

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Meaning of Genetic Imprinting

- Though the male and female genes are responsible for a same trait, the same character from a female differs in behavior compared to that from a male
- Ex. Male sex chromosomes – affect different phenotype compared to females
- Thus, there are genes which are "**Stamped**" or "**IMPRINTED**" sex markers

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Meaning of Genetic Imprinting

- Huntingtons – Autosomal Dominant – the source can be either the father or mother
- In case of paternal origin, time of initiation and severity are more compared to maternal origin
- ***Genetic imprinting is the differential inheritance of genetic material from the mother versus the father***

Mechanism:

- Believed to be DNA Methylation on Cytosine
- Maternal genes are more methylated
- Excessive methylation inhibits functions of the genes

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Case Study

PRADER WILLI SYNDROME

- Deletion in position 11, 12, 13 in Ch. 15
- Always derived paternally
- Development delay, obesity and hypogonadism

ANGELMAN SYNDROME

- A maternally inherited disorder of chromosome deletion 11, 12, 13 in Ch. 15
- Happy disposition, mental retardation, large mouth, protruding tongue and seizures

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Nature of Genetic Imprinting

1. **Imprinted Genes Express Variously**
   - Male and female genes have different magnitudes of expression and time of expression
2. **Genetic Imprints are Erasable**
   - Imprinting done in gonadal tissue - Such markings are erased during gametogenesis - offspring marks his own sex
3. **Genetic Imprinting is Not a Rule**
   - Discovered only in some male and female genes - Majority of genes are not imprinted
4. **Genetic Imprinting is Species Specific**
   - A gene imprinted in one species may not be imprinted in another species

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Genetic Screening

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What is genetic screening?

- One of the fastest moving fields in medical science.
- A technique to determine the genotype or phenotype of an organism.
- It is often used to detect faulty or abnormal genes in an organism.
- Some examples
  - Prenatal screening
  - Newborn screening
  - Carrier screening

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Prenatal Screening

- This can detect a disorder before a baby is born.
- An ultrasound test is used to determine if the fetus is at a high or low risk from a genetic disorder.
- **Amniocentesis is used to extract fetal cells**
- If diagnosed early in the pregnancy, there is still the possibility of abortion.
- Prenatal screening is sometimes seen as controversial.

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Newborn Screening

- Newborns are tested for diseases and early diagnoses allows for immediate treatment.
- A blood sample is tested for genetic disorders.
- In most of the USA, newborn screening is mandatory, unless parents have a religious objection to it.

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Carrier Screening

- This involves testing prospective parents for diseases that they show no symptoms of, but may carry a recessive gene for.
- A blood sample or cheek cell sample is analysed to determine whether either parent carries a faulty gene.
- If both parents carry a specific faulty gene, the chance of the fetus receiving the gene from both parents is 25%, and the chance of being a carrier is 50%.
- If both parents carry a faulty gene, they may decide to have prenatal testing on the fetus.

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Other types of screening

- **Preimplantation screening:** Screening embryos fertilised by IVF before they are implanted into the uterus.
- **Presymptomatic screening:** Screening to predict adult-onset diseases such as Huntington's disease.
- **Presymptomatic screening:** Screening to estimate the risk of developing cancer or Alzheimer's disease as an adult.
- **Forensic/Identity testing:** Screening to eg. determine the father of an individual (paternity test).

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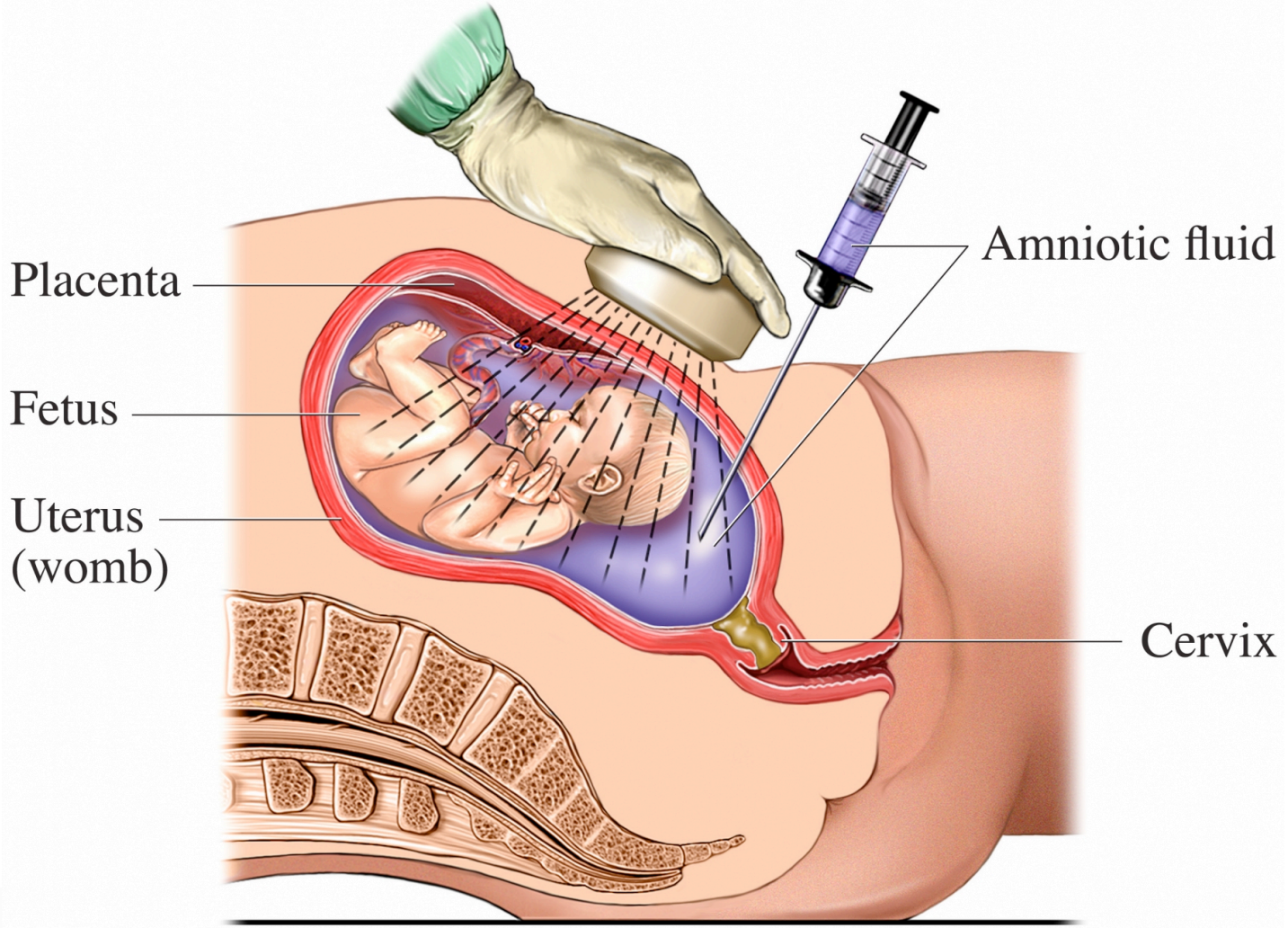
Amniocentesis

- If parents heterozygous for a genetic disease decide to produce a child, it is now possible to advise fi their child has the inherited disease after about two months of conception through amniocentesis
- The cultured fetal cells may be used for determining their karyotype, levels of critical enzymes and restriction patterns of DNA
- This is an **antenatal** diagnosis
- Unfortunately misused for selective abortion of one sex
- Based on the findings, corrective measures can be taken

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Amniocentesis



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Genetic Counseling

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What is Genetic Counseling

- Consists of educating prospective parents either suffering from / suspected of heterozygous for some genetic disease on risk of their children suffering from same disease

Involves

1. Genetic screening of disorder and explanation to parents about the genetic or clinical implications
2. Calculation of likelihood of recurrence of such genetic disorder in the family
3. To suggest ways in which occurrence of genetic disorder can be controlled

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Problems

- Genetic counseling and antenatal diagnosis definitely are a relief to parents and reduce the frequency of genetically defective individuals in a population
- However, there are some issues…
  1. Heterozygozity: Most genetic defects are recessive. Even if there is a total ban on reproduction by homozygotes, they would remain in population through heterozygotes
  2. Mistaken paternity
  3. Delayed counseling
  4. Issues of confidentiality
  5. Screening an asymptote like a relative
  6. Difficulty in counseling late developing diseases

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